

Short multiplication

Multiplying a large number by a single-digit number is called short multiplication. The larger number is placed above the smaller one in columns arranged according to their value.

$$\begin{array}{r} 4 \text{ carried} \\ \text{to tens} \\ \text{column} \end{array} \begin{array}{r} 196 \\ \times 7 \\ \hline \end{array}$$

6 written in ones column

2 written in ones column

$$\begin{array}{r} 6 \text{ carried} \\ \text{to hundreds} \\ \text{column} \end{array} \begin{array}{r} 196 \\ \times 7 \\ \hline \end{array}$$

7 written in tens column

$$\begin{array}{r} 1 \text{ written in} \\ \text{hundreds} \\ \text{column} \end{array} \begin{array}{r} 196 \\ \times 7 \\ \hline \end{array}$$

3 written in hundreds column; 1 written in thousands column

1,372 is final answer

To multiply 196 and 7, first multiply the ones 7 and 6. The product is 42, the 4 of which is carried.

Next, multiply 7 and 9, the product of which is 63. The carried 4 is added to 63 to get 67.

Finally, multiply 7 and 1. Add its product (7) to the carried 6 to get 13, giving a final product of 1,372.

Long multiplication

Multiplying two numbers that both contain at least two digits is called long multiplication. The numbers are placed one above the other, in columns arranged according to their value (ones, tens, hundreds, and so on).

$$\begin{array}{r} 428 \\ \times 111 \\ \hline \end{array}$$

428 multiplied by 1

First, multiply 428 by 1 in the ones column. Work digit by digit from right to left so 8×1 , 2×1 , and then 4×1 .

$$\begin{array}{r} 428 \\ \times 111 \\ \hline \end{array}$$

428 multiplied by 10

add 0 when multiplying by 10

Multiply 428 digit by digit by 1 in the tens column. Remember to add 0 when multiplying by a number in the tens place.

$$\begin{array}{r} 428 \\ \times 111 \\ \hline \end{array}$$

428 multiplied by 100

add 00 when multiplying by 100

Multiply 428 digit by digit by 1 in the hundreds column. Add 00 when multiplying by a digit in the hundreds place.

$$\begin{array}{r} 428 \\ \times 111 \\ \hline 428 \\ + 4,280 \\ + 42,800 \\ \hline = 47,508 \end{array}$$

Add together the products of the three multiplications. The answer is 47,508.

LOOKING CLOSER

Box method of multiplication

The long multiplication of 428 and 111 can be broken down further into simple multiplications with the help of a table or box. Each number is reduced to its hundreds, tens, and ones, and multiplied by the other.

► The final step

Add together the nine multiplications to find the final answer.

428 WRITTEN IN 100S, 10S, AND ONES			
	400	20	8
100	$400 \times 100 = 40,000$	$20 \times 100 = 2,000$	$8 \times 100 = 800$
10	$400 \times 10 = 4,000$	$20 \times 10 = 200$	$8 \times 10 = 80$
1	$400 \times 1 = 400$	$20 \times 1 = 20$	$8 \times 1 = 8$

$$\begin{array}{r} 40,000 \\ 2,000 \\ 800 \\ 4,000 \\ 200 \\ 80 \\ 400 \\ 20 \\ 8 \\ + \\ \hline = 47,508 \end{array}$$

this is the final answer